

Exercise 1: Control Structures

CODE:

```
CREATE TABLE Customers (
```

```
    CustomerID  NUMBER PRIMARY KEY,
```

```
    CustomerName VARCHAR2(50),
```

```
    Age        NUMBER,
```

```
    Balance    NUMBER,
```

```
    IsVIP      VARCHAR2(5)
```

```
);
```

```
CREATE TABLE Loans (
```

```
    LoanID     NUMBER PRIMARY KEY,
```

```
    CustomerID NUMBER,
```

```
    InterestRate NUMBER,
```

```
    DueDate    DATE,
```

```
    CONSTRAINT fk_customer FOREIGN KEY (CustomerID) REFERENCES Customers (CustomerID)
```

```
);
```

```
INSERT INTO Customers VALUES (1, 'Rahul', 65, 15000, 'FALSE');
```

```
INSERT INTO Customers VALUES (2, 'Priya', 45, 8000, 'FALSE');
```

```
INSERT INTO Customers VALUES (3, 'Rohan', 70, 12000, 'FALSE');
```

```
INSERT INTO Customers VALUES (4, 'Anita', 30, 5000, 'FALSE');
```

```
INSERT INTO Loans VALUES (101, 1, 10, SYSDATE + 20);
```

```
INSERT INTO Loans VALUES (102, 2, 8, SYSDATE + 50);
```

```
INSERT INTO Loans VALUES (103, 3, 9, SYSDATE + 10);
```

```
INSERT INTO Loans VALUES (104, 4, 11, SYSDATE + 100);
```

```
COMMIT;
```

```
-- Scenario 1: Reduce interest rate by 1 for customers older than 60
```

```
SET SERVEROUTPUT ON;
```

```
DECLARE
```

```
BEGIN
```

```
    FOR c IN (SELECT CustomerID, Age FROM Customers)
```

```
    LOOP
```

```
        IF c.Age > 60 THEN
```

```
            UPDATE Loans
```

```
                SET InterestRate = InterestRate - 1
```

```
                WHERE CustomerID = c.CustomerID;
```

```
        END IF;
```

```
    END LOOP;
```

```
COMMIT;
```

```
DBMS_OUTPUT.PUT_LINE('Interest rates updated.');
```

```
END;
```

```
/
```

```
SELECT * FROM Loans;
```

```
-- Scenario 2: Mark customers with balance > 10000 as VIP
```

```
SET SERVEROUTPUT ON;
```

```
BEGIN
```

```

FOR c IN (SELECT CustomerID, Balance FROM Customers)
LOOP
    IF c.Balance > 10000 THEN
        UPDATE Customers
        SET IsVIP = 'TRUE'
        WHERE CustomerID = c.CustomerID;
    END IF;
END LOOP;
COMMIT;
DBMS_OUTPUT.PUT_LINE('VIP status updated.');
```

END;

/


```

SELECT * FROM Customers;
```


-- Scenario 3: Send reminders for loans due within the next 30 days

```

SET SERVEROUTPUT ON;
BEGIN
    FOR l IN (
        SELECT c.CustomerName, l.DueDate
        FROM Customers c
        JOIN Loans l ON c.CustomerID = l.CustomerID
        WHERE l.DueDate <= SYSDATE + 30
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE(
```

```

        'Reminder: '
        || I.CustomerName
        || ' has a loan due on '
        || TO_CHAR(I.DueDate, 'DD-MON-YYYY')
    );
END LOOP;

END;

/

```

OUTPUT:

All rows fetched: 4 in 0.161 seconds

	LOANID	CUSTOMERID	INTERESTRATE	DUEDATE
1	101	1	8	20-07-26
2	102	2	8	19-08-26
3	103	3	7	10-07-26
4	104	4	11	08-10-26

All rows fetched: 4 in 0.162 seconds

	CUSTOMERID	CUSTOMERNAME	AGE	BALANCE	ISVIP
1	1	Rahul	65	15000	TRUE
2	2	Priya	45	8000	FALSE
3	3	Rohan	70	12000	TRUE
4	4	Anita	30	5000	FALSE

```

Reminder: Rahul has a loan due on 20-JUL-2026
Reminder: Rohan has a loan due on 10-JUL-2026

```

```

PL/SQL procedure successfully completed.

```

```

Reminder: Rahul has a loan due on 20-JUL-2026
Reminder: Rohan has a loan due on 10-JUL-2026

```